

Anastasiia Tkachenko

Computational Geometry | Algorithms | Theoretical Computer Science

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RESEARCH INTERESTS

Computational geometry and graph algorithms, with emphasis on geometric intersection graphs and optimization problems for points and disks in convex position. Current topics include dominating set, independent set, clique, dispersion, and k-center problems.

EDUCATION

Ph.D. in Computer Science (expected May 2027), The University of Utah **2023-Present**

Advisor: Professor Haitao Wang

GPA: 4.0

Doctoral Coursework in Computer Science, Utah State University **2021-2023**

GPA: 4.0

M.S. in Applied Mathematics and Informatics, Novosibirsk State University **2016-2018**

Diploma with highest honors (summa cum laude)

Thesis: Maximization of labor productivity in a two-level optimal control problem: perturbed production function

B.S. in Applied Mathematics and Informatics, Novosibirsk State University **2012-2016**

Diploma with highest honors (summa cum laude)

PUBLICATIONS • COMPUTATIONAL GEOMETRY

Journal Article

1. **Anastasiia Tkachenko** and Haitao Wang. "Computing Maximum Cliques in Unit Disk Graphs." *Computational Geometry: Theory and Applications*, 135, Article 102269, pp. 1-12, 2026. [DOI](#).

Conference Papers

1. **Anastasiia Tkachenko** and Haitao Wang. "Maximum Independent Sets in Disk Graphs with Disks in Convex Position." *Proceedings of the 20th Scandinavian Symposium on Algorithm Theory (SWAT 2026)*, pp. 40:1-40:18, 2026. *Presented by Anastasiia Tkachenko*. [Paper](#).
2. **Anastasiia Tkachenko** and Haitao Wang. "Computing Dominating Sets in Disk Graphs with Centers in Convex Position." *Proceedings of the 17th Latin American Theoretical Informatics Symposium (LATIN 2026)*, 2026. *Presented by Anastasiia Tkachenko*. [arXiv](#).
3. **Anastasiia Tkachenko** and Haitao Wang. "Computing Maximum Cliques in Unit Disk Graphs." *Proceedings of the 37th Canadian Conference on Computational Geometry (CCCG 2025)*, pp. 283-291, 2025. [Paper](#).
4. **Anastasiia Tkachenko** and Haitao Wang. "Dominating Set, Independent Set, Discrete k-Center, Dispersion, and Related Problems for Planar Points in Convex Position." *Proceedings of the 42nd International Symposium on Theoretical Aspects of Computer Science (STACS 2025)*, pp. 73:1-73:20, 2025. *Presented by Anastasiia Tkachenko*. [Paper](#).

Manuscripts Under Review

1. **Anastasiia Tkachenko** and Haitao Wang. "Maximum Independent Sets in Disk Graphs with Disks in Convex Position." Submitted to *Journal of Computational Geometry (JoCG)*; *under review*. [Preprint](#).

2. **Anastasiia Tkachenko** and Haitao Wang. “Dominating Set, Independent Set, Discrete k-Center, Dispersion, and Related Problems for Planar Points in Convex Position.” Submitted to *Algorithmica*; *under review*. [Preprint](#).
3. **Anastasiia Tkachenko** and Haitao Wang. “Computing Dominating Sets in Disk Graphs with Centers in Convex Position.” Submitted to *Computational Geometry: Theory and Applications (CGTA)*; *major revision*. [Preprint](#).
4. **Anastasiia Tkachenko** and Haitao Wang. “Computing Independent Sets of Size Three in Disk Graphs.” Submitted to the 21st International Conference and Workshops on Algorithms and Computation (WALCOM); *under review*.

Manuscript in Preparation

1. **Anastasiia Tkachenko** and Haitao Wang. “The Clique Problem in Disk Graphs: Deterministic FPT and Convex-Position Algorithms.” Planned submission to the 43rd International Symposium on Computational Geometry (SoCG).

INTERDISCIPLINARY PUBLICATIONS

Journal Articles

1. H. Wang, J. B. Brown, C. Council, **A. Tkachenko**, Y. Qian, L. Xue, Z. Li, X. Qi, and Y. Rao. “Theory and Experiments of Angle-Dependent Second-Order Nonlinear Optical Scattering.” *Physical Review B*, 2026. [DOI](#).
2. N. Tkachenko, **A. A. Tkachenko**, B. Nebgen, S. Tretiak, and A. I. Boldyrev. “Neural Network Atomistic Potentials for Global Energy Minima Search in Carbon Clusters.” *Physical Chemistry Chemical Physics*, 25, 21173-21182, 2023. [DOI](#).
3. V. A. Kulyukin, D. Coster, **A. Tkachenko**, D. Hornberger, and A. V. Kulyukin. “Ambient Electromagnetic Radiation as a Predictor of Honey Bee (*Apis mellifera*) Traffic in Linear and Non-Linear Regression: Numerical Stability, Physical Time and Energy Efficiency.” *Sensors*, 23, 2584, 2023. [DOI](#).
4. V. A. Kulyukin, **A. A. Tkachenko**, K. Price, W. Meikle, and M. Weiss. “Integration of Scales and Cameras in Nondisruptive Electronic Beehive Monitoring: On the Within-Day Relationship of Hive Weight and Traffic in Honeybee (*Apis mellifera*) Colonies in Langstroth Hives in Tucson, Arizona, USA.” *Sensors*, 22, 4824, 2022. [DOI](#).
5. N. V. Tkachenko, **A. A. Tkachenko**, V. A. Kulyukin, and A. I. Boldyrev. “DFT Study of Microsolvated $[\text{NO}_3 \cdot (\text{H}_2\text{O})_n]^-$ ($n = 1-12$) Clusters and Molecular Dynamics Simulation of Nitrate Solution.” *The Journal of Physical Chemistry A*, 125, 8899, 2021. [DOI](#).
6. V. A. Kulyukin, N. Ganta, and **A. A. Tkachenko**. “On Image Classification in Video Analysis of Omnidirectional *Apis Mellifera* Traffic: Random Reinforced Forests vs. Shallow Convolutional Networks.” *Applied Sciences*, 11(17), 8141, 2021. [DOI](#).

WORKSHOP AND FORUM PRESENTATIONS

1. **Anastasiia Tkachenko** and Haitao Wang. “Maximum Independent Sets in Disk Graphs with Disks in Convex Position.” *Computational Geometry: Young Researchers Forum (CG:YRF)*, CG Week 2026, Rutgers University, New Brunswick, New Jersey, June 2-5, 2026. [Program](#).
2. **Anastasiia Tkachenko** and Haitao Wang. “Computing Dominating Sets in Disk Graphs with Centers in Convex Position.” *32nd Annual Fall Workshop on Computational Geometry (FWCG 2025)*, Queens, New York, November 7-8, 2025. [Program](#).
3. **Anastasiia Tkachenko** and Haitao Wang. “Independent Set and Dispersion Problems for Planar Points in Convex Position.” *31st Annual Fall Workshop on Computational Geometry (FWCG 2024)*, Tufts University, Medford, Massachusetts, November 15-16, 2024. [Program](#).

RESEARCH EXPERIENCE

- Research Assistant**, Kahlert School of Computing, The University of Utah **2023-2026**
- Designed and analyzed algorithms for domination, independent set, clique, dispersion, and k-center problems in geometric intersection graphs.
 - Developed structural results for points and disks in convex position that enable efficient algorithms.
 - Research resulted in four peer-reviewed conference papers, one peer-reviewed journal article, and three workshop/forum presentations.
- Research Assistant**, Utah State University **May-Aug 2023**
- Departments:** Computer Science; Chemistry and Biochemistry
- Developed the theoretical framework for sum-frequency-generation scattering spectroscopy and prepared manuscript materials.
- Research Assistant**, Novosibirsk State University **2016-2018**
- Developed mathematical models and established their validity and applicability, conducted literature reviews and prepared materials for grant submissions on mathematical models of government-investor economic processes.
 - Research resulted in four conference papers and abstracts.

TEACHING EXPERIENCE

- Teaching Assistant**, The University of Utah **2024-Present**
- Computational Geometry, Teaching Assistant, Fall 2026
 - Algorithms (CS 4150), Lead Teaching Assistant, Spring 2026; Teaching Assistant, Spring 2024
 - Models of Computation (CS 3100), Lead Teaching Assistant, Spring 2025
- Led discussion sections; graded homework, programming assignments, exams, and projects; supervised and coordinated the teaching-assistant team as lead TA.
- Teaching Assistant**, Utah State University **2021-2023**
- Programming Languages (CS 4700), delivered several course lectures, Spring 2023
 - Advanced Algorithms (CS 5050), delivered several course lectures, Fall 2022
 - Scientific Computing (CS 3430), Spring 2022
 - Theory of Computability (CS 5000), Fall 2021
 - Introduction to Computer Science (CS 1400), Spring 2021
- Prepared lessons and instructional materials, facilitated discussion sections, and graded coursework and exams

HONORS AND GRANTS

- Graduate Student Professional Development Supplement Award**, **2026**
The University of Utah (\$500)
- CG Young Researchers Forum Travel Grant**, Rutgers University (\$500) **2026**
- Fall Workshop on Computational Geometry Travel Grant**, Tufts University (\$1,000) **2024**
- Outstanding Computer Science Graduate Teaching Assistant**, **2023**
Utah State University (\$100)
- Graduate Student Enhancement Scholarship**, Utah State University (\$4000) **2022-2023**
- Personal Fellowship of Sberbank - 12UP**, National competition (\$3000) **2017-2018**
- Second-Degree Diploma**, International Scientific Student Conference **2017**
- Scholarship for Scientific Achievements**, Novosibirsk State University (\$1500) **2017**

Participant, The Russian Foundation for Basic Research Grant:	2016-2018
Mathematical analysis of dynamic models of non-classical markets (\$2500)	
Participant, The Russian Foundation for Basic Research Grant:	2016-2018
Analysis of complex socio-economic systems: new approaches and methods (\$1000)	

SERVICE AND LEADERSHIP

Peer Reviewer , International Symposium on Computational Geometry (SoCG); European Symposium on Algorithms (ESA); Engineering Optimization	2025-2026
Conference Volunteer , ACM STOC 2026, Salt Lake City, Utah	June 2026
President , Computer Science Graduate Student Council, Utah State University	2022-2023
Vice President , Association for Computing Machinery for Women, Utah State University	2021-2023

ADDITIONAL CONFERENCE ABSTRACTS

Selected earlier publications appeared under the name A. A. Kotova.

- A. A. Tkachenko.** "On a Special Rational Control Mode with Perturbation of the Production Function." *Book of Abstracts of the 56th International Scientific Student Conference*, p. 105, 2018.
- A. A. Kotova.** "On the Influence of Investors' Propensity to Consume on the Rate of Personal Income Tax." *Book of Abstracts of the International Conference Dedicated to the 60th Anniversary of the Sobolev Institute, Mathematics in the Modern World*, p. 544, 2017.
- A. A. Kotova.** "On Maximizing Labor Productivity in a Two-Level Optimal Control Problem." *Book of Abstracts of the 55th International Scientific Student Conference*, p. 197, 2017.
- A. A. Kotova.** "On Maximizing Labor Productivity in the Modified Ramsey-Solow Model." *Book of Abstracts of the 54th International Scientific Student Conference*, p. 215, 2016.